

COVID-19 Global Data Analysis Report

1. Executive Summary

This project presents an end-to-end analysis of global COVID-19 data using advanced SQL techniques and Power BI visualizations. The objective was to uncover patterns related to the severity, spread, and progression of the pandemic across countries and regions.

Key insights from the analysis show that countries with high Case Fatality Rates often reflect either healthcare system stress or underreporting of cases. Infection rates relative to population provide a more accurate measure of disease penetration than absolute case counts. The pandemic demonstrated synchronized global waves, where multiple countries experienced peaks around the same periods. Recovery trends consistently lagged behind confirmed cases, indicating delays in healthcare response and reporting systems. Additionally, several inconsistencies in reporting highlight data quality challenges across regions.

2. Problem Statement

The objective of this project was to perform a comprehensive analysis of COVID-19 data using SQL in order to:

- Measure the severity of the pandemic
- Understand its spread across populations
- Analyze temporal trends and growth patterns
- Identify inconsistencies in reported data

3. Tools and Technologies

- SQL (advanced queries, window functions, aggregations)
- Power BI (data visualization and dashboarding)
- Dataset: Global COVID-19 dataset

4. Data Cleaning and Preprocessing

Data preprocessing was a critical step to ensure accuracy and reliability of the analysis. The following actions were performed:

- Handled missing values in key columns such as confirmed cases, deaths, and recoveries
- Standardized date formats for time-series analysis
- Removed duplicate and inconsistent records
- Verified cumulative metrics to maintain logical consistency

5. Analysis and Findings

5.1 Case Fatality Rate (CFR)

Objective: Measure the severity of COVID-19 across countries by calculating deaths as a percentage of confirmed cases.

Method: CFR was calculated using the formula: $(\text{Total Deaths} / \text{Total Confirmed Cases}) \times 100$. Countries were then ranked based on CFR values.

Findings: The highest CFR values were observed in Vanuatu, Yemen, Peru, Mexico, and Sudan. Lower CFR values were recorded in Singapore, Bhutan, Laos, Burundi, and Timor-Leste.

Interpretation: A high CFR does not necessarily indicate a more severe outbreak. It may reflect underreporting of cases, limited testing capacity, delayed diagnosis, or healthcare system strain. This metric is useful for comparing relative severity across countries.

	country_region	percent_f_death_to_confirm
►	Vanuatu	25.00%
	Yemen	19.66%
	Peru	9.42%
	Mexico	9.38%
	Sudan	7.55%
	Syria	7.29%
	Egypt	5.72%
	Somalia	5.23%
	China	4.70%
	Ecuador	4.68%

5.2 Infection Rate versus Population

Objective: Identify countries where COVID-19 spread had the greatest impact relative to population size.

Method: Infection rate was calculated by comparing total confirmed cases with total population, and countries were ranked based on infection penetration.

Findings: The highest infection rates were observed in Andorra, Bahrain, Montenegro, Czechia, and San Marino. Other significantly affected countries included Maldives, Seychelles, Slovenia, Luxembourg, and Sweden.

Interpretation: Normalizing cases by population provides a clearer understanding of spread. Smaller countries often appear disproportionately affected due to higher population penetration despite lower absolute case counts.

	country_region	infection_rate
►	Andorra	0.1785
	Bahrain	0.1757
	Montenegro	0.1595
	Czechia	0.1559
	San Marino	0.1472
	Maldives	0.1400
	Seychelles	0.1318
	Slovenia	0.1220
	Luxembourg	0.1138
	Sweden	0.1046

5.3 Regional Contribution to Deaths

Objective: Determine which regions contributed the most to national death totals in countries with multiple administrative divisions.

Method: Deaths were grouped by country and region, and each region's contribution was compared against the national total.

Findings: Countries such as Denmark, the United Kingdom, France, the Netherlands, and China showed highly concentrated regional death distributions, often dominated by a single region.

Interpretation: In many countries, deaths were concentrated in specific regions, indicating localized outbreak severity. In contrast, some countries exhibited more distributed impacts across regions.

	country_region	province	countries_total_deaths	province_total_deaths	percentage_contribution
►	Denmark	Denmark	2532	2531	99.96
	United Kingdom	United Kingdom	128171	127904	99.79
	France	France	110966	109913	99.05
	Netherlands	Netherlands	18029	17750	98.45
	China	Hubei	4829	4495	93.08
	Australia	Victoria	911	820	90.01
	Canada	Quebec	25924	11173	43.10

5.4 Time to First 100 Deaths

Objective: Measure how quickly countries reached 100 cumulative deaths after the first reported case.

Method: The time difference between the first confirmed case and the date of reaching 100 deaths was calculated for each country.

Findings: China, Tajikistan, Iran, Turkey, and Switzerland showed the fastest progression. The average duration across countries was approximately 109 days.

Interpretation: Faster progression indicates rapid outbreak escalation, while longer durations suggest slower spread or more effective early containment measures.

	country_region	days
▶	China	2
	Tajikistan	4
	Iran	8
	Turkey	9
	Switzerland	10
	Qatar	11
	Denmark	12
	Panama	12
	Netherlands	13
	Norway	13

5.5 Global Hotspot Center

Objective: Identify the geographic center of the most severe outbreak zones using the top 5 percent of hotspot locations weighted by case counts.

Method: A weighted average of latitude and longitude was calculated for the highest case concentration regions.

Findings: The weighted center was approximately at latitude 22.28 and longitude -14.66, with total cases exceeding 113 million in top hotspot regions.

Interpretation: The results indicate that COVID-19 severity was geographically clustered, influenced by travel connectivity, population density, and transmission dynamics.

	weighted_center_lat	weighted_center_long	total_cases_in_top_5_percent
▶	22.2791162289362311	-14.6614879432135000	113821119

6. Advanced Temporal and Statistical Analysis

6.1 Global Recovery Lag

Objective: Measure the average time delay between global surges in confirmed cases and subsequent peaks in reported recoveries.

Method: The peak date for global confirmed cases was compared against the peak date for global recoveries using the DATEDIFF function applied to aggregated daily totals.

Findings: The global peak for confirmed cases occurred on December 10, 2020, while the peak for recoveries followed on December 12, 2020. This resulted in a statistical recovery lag of only 2 days.

Interpretation: Biologically, COVID-19 recovery typically spans 10–14 days. The 2-day result suggests that recovery data was often not reported in real-time but rather "batched" or back-filled administratively, causing recovery peaks to appear artificially close to infection peaks.

	date_of_max_confirmed	date_of_max_recovery	days_from_peak_cases_to_most_recovery
▶	2020-12-10	2020-12-12	2

6.2 Data Integrity Audit

Objective: Identify countries with reporting inconsistencies where the combined total of deaths and recoveries exceeds the total number of confirmed cases.

Method: A comparative filter was applied to sum all statuses (Recovered + Deaths) and subtract the Total Confirmed count to identify and rank the magnitude of discrepancies.

Findings: Significant data anomalies were found in Peru (150,026 extra recoveries), Lithuania (22,677), and Luxembourg (3,714).

Interpretation: These discrepancies prove that many nations updated their recovery figures in bulk (back-filling) without retrospective adjustments to confirmed cases. This explains the "2-day lag" mystery in Section 5.1; massive administrative data dumps create artificial peaks that do not reflect the true biological timeline of the virus.

	country_region	total_confirmed	total_deaths	total_recovered	discrepancy
▶	Peru	2003625	188708	1964943	150026
	Lithuania	277746	4339	296084	22677
	Luxembourg	71754	820	74648	3714
	El Salvador	75443	2295	74323	1175
	China	102847	4829	99072	1054
	Cote d'Ivoire	47696	306	48401	1011
	Tajikistan	13308	91	13389	172
	Madagascar	41940	887	41112	59

6.3 Trend Smoothing (7-Day Moving Average)

Objective: Account for daily reporting fluctuations and the "weekend effect" to identify the true trajectory of the pandemic in high-impact nations.

Method: A 7-day moving average was calculated using a window function (ROWS BETWEEN 6 PRECEDING AND CURRENT ROW) for the USA and India.

Findings: For the USA, while daily spikes were erratic, the 7-day average revealed a sustained plateau in mid-April 2021, hovering between 68,000 and 71,000 daily cases.

Interpretation: Moving averages cut through the noise of administrative delays. They demonstrate that even when daily reports showed a "dip" (often on Sundays/Mondays), the viral momentum remained high, providing a more accurate metric for healthcare resource planning.

	country_region	date_raw	seven_day_moving_avg
	India	2020-02-28	0
	India	2020-02-29	0
	India	2020-03-01	0
	India	2020-03-02	0
	India	2020-03-03	0
	India	2020-03-04	4
	India	2020-03-05	4
	India	2020-03-06	4
	India	2020-03-07	4
	India	2020-03-08	5

6.4 Peak Momentum Identification

Objective: Identify the single highest surge date for each country to determine if the pandemic experienced simultaneous global peaks.

Method: Daily case totals were ranked per country using the RANK() window function to isolate the date of maximum intensity.

Findings:

- Turkey: Peaked on 2020-12-10 with 823,225 cases (indicative of a massive data dump).
- India: Peaked on 2021-05-06 with 414,188 cases.
- USA: Peaked on 2021-01-02 with 300,462 cases.

Interpretation: The data shows a "rolling" global peak. The pandemic did not hit everyone at once; momentum shifted from the West (USA peak in Jan) to South Asia (India peak in May), illustrating the waves of transmission across continents.

	country_region	date_raw	cases_per_day
►	Turkey	2020-12-10	823225
	India	2021-05-06	414188
	USA	2021-01-02	300462
	France	2021-04-11	117900
	Brazil	2021-03-25	100158
	Spain	2021-01-25	93822
	United Kingdom	2021-01-08	68192
	Germany	2020-12-30	49044
	Argentina	2021-05-27	41080
	Italy	2020-11-13	40902
	Poland	2021-04-01	35253
	Sweden	2020-12-29	32485
	Colombia	2021-06-10	31656

6.5 Growth Acceleration (Doubling Rate)

Objective: Measure the velocity of the virus during its early phase by calculating how many days it took for total cases to double.

Method: Starting from a threshold of 100 cases, the time difference was calculated between the base date and the date when the cumulative "snowball" of cases reached 200%.

Findings: China experienced the fastest doubling rate of just 1 day. Italy, Argentina, and Denmark followed closely with a doubling rate of 2 days.

Interpretation: A 1-to-2-day doubling rate indicates extreme exponential growth. This explains why healthcare systems in Europe and Asia were overwhelmed so rapidly; the patient load was doubling faster than administrative or medical responses could be implemented.

	country_region	base_date	final_date	number_of_days
▶	China	2020-01-24	2020-01-25	1
	Argentina	2020-03-20	2020-03-22	2
	Benin	2020-05-07	2020-05-09	2
	Chile	2020-03-15	2020-03-17	2
	Denmark	2020-03-10	2020-03-12	2
	Dominican Republic	2020-03-21	2020-03-23	2
	Ecuador	2020-03-18	2020-03-20	2
	Iran	2020-02-26	2020-02-28	2
	Italy	2020-02-23	2020-02-25	2
	Kazakhstan	2020-03-26	2020-03-28	2
	Laos	2021-04-23	2021-04-25	2
	Luxembourg	2020-03-17	2020-03-19	2

6.6 Month-over-Month Growth (2020)

Objective: Track the percentage change in infection levels throughout the first year of the pandemic.

Method: Monthly case totals were compared to the previous month using the LAG function to calculate the percentage growth rate.

Findings: The highest growth rate occurred in March 2020 (938.06% increase) and February (711.08%). By late 2020, while the volume was higher, the growth rate had stabilized to approximately 16%–42%.

Interpretation: The early "percentage" explosions reflect the virus entering a naive population. The late-year growth (Oct–Nov) reflects a "second wave" where the raw number of infections was much higher, even though the percentage increase from the previous month was smaller.

	months	current_cases	previous_cases	percent_change
►	March	789828	76087	938.06%
	February	76087	9381	711.08%
	April	2438024	789828	208.68%
	July	7146948	4274269	67.21%
	June	4274269	2912906	46.74%
	November	17315980	12130638	42.75%
	October	12130638	8510747	42.53%
	May	2912906	2438024	19.48%
	December	20149435	17315980	16.36%
	August	7905912	7146948	10.62%
	September	8510747	7905912	7.65%
	January	9381	NULL	NULL

6.7 Sustained Decline Analysis

Objective: Identify countries that successfully achieved a "continuous decline," defined as 14 or more consecutive days of decreasing new cases.

Method: A "Group ID" logic was used to identify streaks of consecutive days where `daily_cases < previous_day_cases`.

Findings: Algeria achieved the most significant sustained decline with a 32-day streak (Aug 28 – Sept 28, 2020). El Salvador maintained a 23-day streak, while Egypt and Qatar achieved 15–16 days.

Interpretation: A 14-day decline is a key epidemiological indicator that containment measures (lockdowns, masking, or vaccinations) are effectively breaking the chain of transmission. Algeria's 32-day streak stands as a benchmark for successful "curve flattening" in this dataset.

	country_region	streak_start_date	streak_end_date	streak_length
►	El Salvador	2020-08-11	2020-09-02	23
	Algeria	2020-08-28	2020-09-28	32
	Qatar	2021-04-20	2021-05-04	15
	Egypt	2021-05-29	2021-06-13	16

7. Key Insights and Synthesis

The analysis reveals that COVID-19 spread was characterized by exponential growth phases, regional clustering, and inconsistent reporting practices. While some countries experienced rapid outbreaks, others demonstrated effective containment. Data inconsistencies highlight the importance of preprocessing and validation in large-scale epidemiological datasets. Moving averages and trend smoothing techniques were essential in revealing the true trajectory of the pandemic.

8. Limitations

- Variability in reporting standards across countries
- Underreporting of cases in certain regions
- Reporting delays affecting temporal accuracy
- Differences in testing capacity and health policies

9. Conclusion

This project demonstrates that COVID-19 was both a global health crisis and a complex data engineering challenge. The analysis highlights variations in reporting accuracy, healthcare response effectiveness, and spread dynamics. SQL-based analysis combined with visualization tools enabled meaningful insights into the pandemic's progression.

10. Future Improvements

- Integration of vaccination data for deeper insights
- Predictive modeling using machine learning techniques
- Real-time dashboard implementation
- More granular regional-level analysis